CHILLER #3a & 3b SEQUENCE OF OPERATION

The Chiller system will occupy one of two states:

1. "CHLR" (mechanical cooling) mode,
2. "OFF" mode A Boolean point will be defined to set the run or stop state of the Chiller and its associated CHW pump, CW pump, and cooling towers.

This Boolean point will be under automatic control, or set to manual override by the operator.

Chillers 3a and 3b will be controlled and staged by the Carrier Chiller Vu 23XRV Series Counterflow Controller. The software in this controller will stage the series counterflow chillers for optimum operation. The software shall at a minimum provide the following:

1. Provide chiller staging, fault logic, and automatically start and stop chillers.
2. Provide machine-specific sequencing to provide the correct online chiller capacity at any given time.
3. Provide soft start ramp loading and chilled water reset.
4. Balance the load efficiently between the two machines.

When indexing from "OFF" mode to "CHLR" mode:

1. The chilled water pump will be started and ramped up to its setpoint flow rate over a two minute period. The primary CHW pump speed will be controlled by the BAS.
2. The condenser water pump will be started. (At minimum speed setting to maintain chiller condenser water allowable minimum flow).
3. Once the chilled water and condenser water pumps have proven that they are running the chiller will be enabled to run.

When indexing from "CHLR" mode to "OFF" mode:

1. The chillers will be commanded off.
2. After the chillers are off, the chilled water pump will run for a minimum of three minute (adj.) before being commanded off.
3. After the chillers are off, the condenser water pump will run for a minimum of two minutes (adj.) before being commanded off. The condenser water pump will stop only after the associated cooling tower fans stop to avoid overloading the tower fans and stress the mechanical drive system.
4. The cooling tower fans will stop.

Cooling Tower CT-3 and condenser water flow control

Condenser Water Pump Control

1. Refer to the sequence of operation on sheets I.106.8 and I.106.9 for cooling tower and condenser water operation and control.
2. Condenser water pump hand-off-auto operation: HOA settings shall be provided as part of the variable frequency drive through the drive's keypad. In the off mode, the pump shall be stopped. In the hand mode, the pump shall run continuously. In the auto mode, the pump shall be started and stopped by the BAS.
3. Condenser water pump local-remote speed control: local-remote settings shall be provided as part of each variable frequency drive through the drive's keypad. In the local mode, the pump's speed shall be controlled through a manual speed control located at the respective drive control panel. In the remote mode, the pump's speed shall be controlled by the BAS.
4. When the chiller is commanded to start, the chiller's condenser control isolation valve and evaporator control isolation valve are opened fully (if applicable). The tower bypass control valve is positioned (100% open) to allow water flow through the tower fill. If outside air temperature is below 45°F (adj.) the cooling tower and condenser water flow control shall be placed in Freeze Protection Mode. Refer to pertinent section for sequence of operations in this mode.
5. Following a time delay, the condenser water pump is started and placed under speed control through its variable speed drive, initially to maintain the minimum system flow rate. The pump VFD setting for minimum allowable flow shall be established during commissioning. The minimum flow set point, for standard chiller operation, is established as the lowest value of the minimum allowable chiller condenser water flow and minimum allowable flow rate for the operational cooling towers. The flow set point, for standard chiller operation, is established as the chiller's full condenser design flow rate for the chiller commanded to run.
6. If the cooling tower pump fails to start, as indicated by its current sensor and its VFD power output, the pump shall be stopped. The condition shall be reported through the BAS with a message that pump service is required.
7. The condenser water supply set-point will be calculated by the BMS based on the fixed approach method. The temperature difference between the condenser water leaving the cooling tower and the ambient wet-bulb temperature shall be fixed at 7°F (fixed approach control). The minimum safe temperature shall be equal to 65°F(adj.) and the maximum safe temperature shall be 85°F (adj.). Refer to Sheet I-106.8 for direction on the Fixed Approach Method.
8. An alternate water supply set-point option will be provided. A near optimal control option will be user selectable for each chiller. Refer to Sheet I-106.8 for direction on the Near Optimal Control Option.

US Army Corps of Engineers
HUNTSVILLE CENTER
HUNTSVILLE, AL

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MARK	DESCRIPTION	DATE	APPR
1	TDP-1 COMMENT REVISIONS	02/16/18	
2	TDP-1 COMMENT REVISIONS 2	07/18/18	

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PLOT SCALE:
ISSUE DATE: 07-06-18
SIZE: ANSI D
FILE NAME:
DATE: 20 NOVEMBER 2017
SOLICITATION NO.: N/A
CONTRACT NO.: W912DY-15-D-0009-0010
FILE NUMBER:

MULTIPLE UMCS PROJECT ALC
ADELPHI LABORATORY CENTER, ADELPHI, MD
W912DY-15-D-0009-0010

BUILDING 106 - CHILLER #3 SEQUENCE OF OPERATION

SHEET IDENTIFICATION
I-106.31
SHEET 21 OF 44